

DATA-DRIVEN CREDIT LINE MANAGEMENT A SEGMENTATION APPROACH



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Abstract

This project uses unsupervised machine learning, specifically K-Means and Hierarchical Clustering, to segment credit card customers based on usage, spending, and payment behaviors. By analyzing credit card transaction data, the study aims to develop personalized, data-driven credit management strategies.

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PROJECT INFRASTRUCTURE & SCALE

8,950
CUSTOMERS

17
FEATURES

4
PCAS

64.1%
EXP VAR

- Data Volume : Analyzed 17 distinct behavioral metrics
- Dimensionality Complexity :
- Granular Segmentation :
 1. Strategic Level : 4 cluster (K Means Model)
 2. Operational Level : 5 cluster (Hierarchical Model)

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BUSINESS QUESTIONS

BQ1 - Who qualifies for higher limits?

BQ2 - Who are the best candidates for elite cards ?

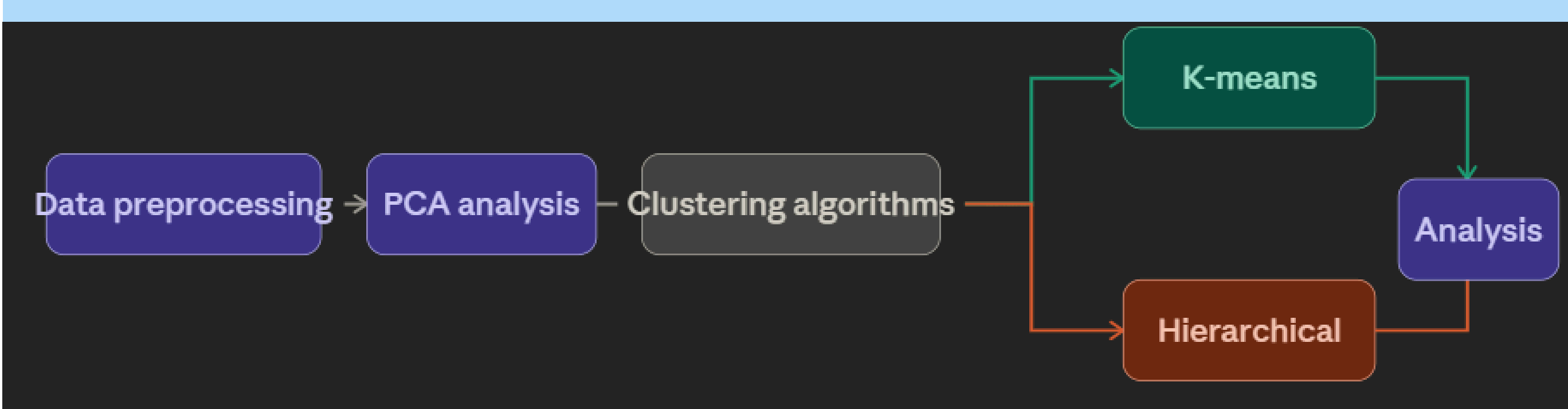
BQ3 - Which segments require limit reductions?

BQ4 - How can we personalize products by behavior?

03

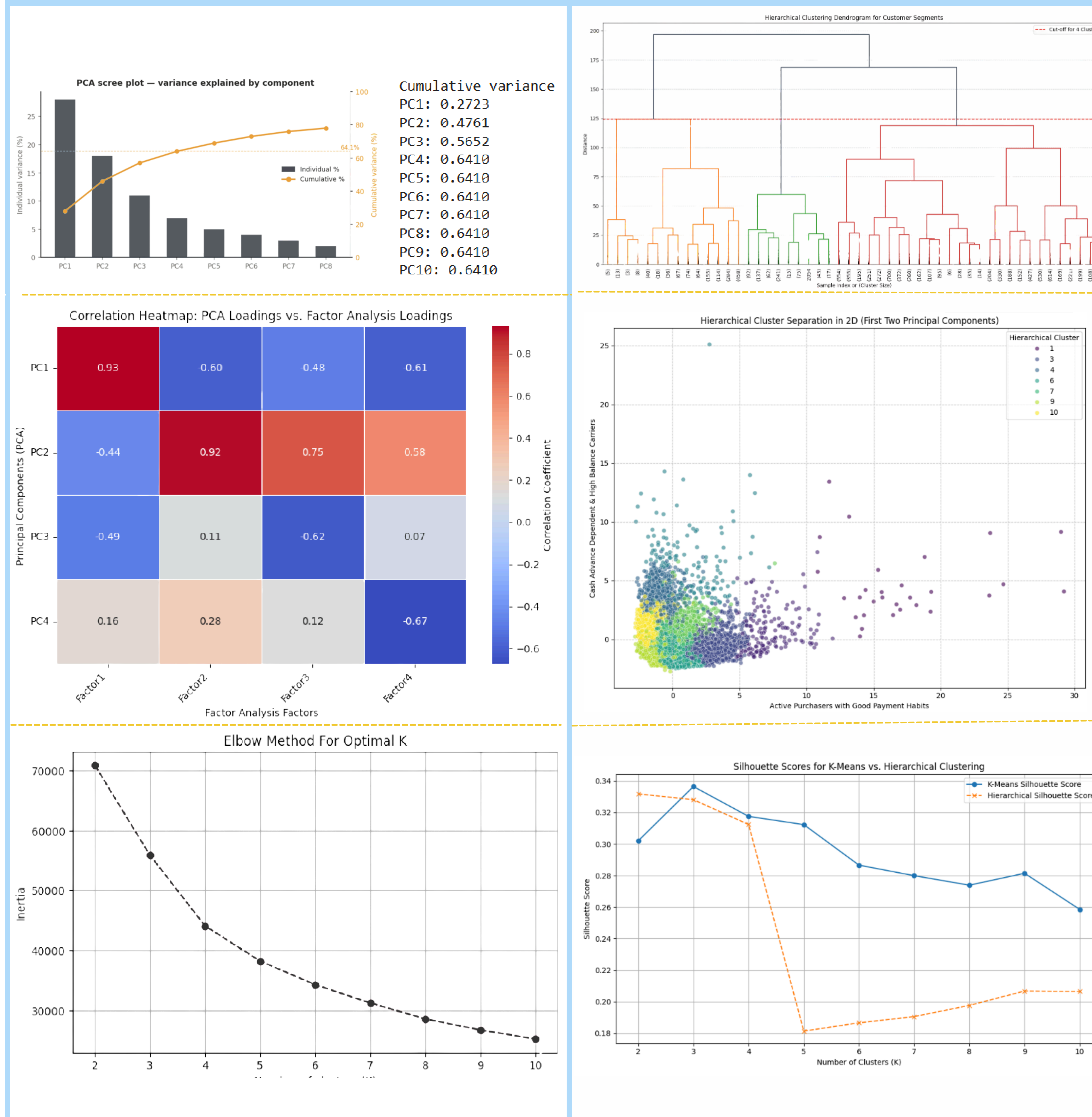
METHODOLOGY

- Data Preprocessing
- Dimensionality Reduction (PCA) / Factor loading
- Clustering Algorithms (K means / Hierarchical)
- Silhoutte Scores for K-Means & Hierachical Clustering



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VISUALISATIONS



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ANALYSIS

- PC1: Active purchasers with good payment habits.
- PC2: Cash advancement dependents and high balance carriers .
- PC3: Long-term, balance-cautious customers.
- PC4: Prudent, small-scale credit utilizers
- Justification of 4 clusters

Silhouette Score : K-means (0.3175)

Silhouette Score : Hierarchical (0.3124)

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RECOMMENDATIONS

Segment	Share	Risk	Limit action	Recommended offer
Frequent purchasers	36.5%	Low	Increase	Premium upgrades, rewards
Low engagement	44.1%	Low	Review	Re-engagement campaigns
Cash advance dependents	13.9%	High	Reduce	Counselling, no marketing
High-value spenders	5.5%	Med	Maintain	Balance transfer, concierge

- CL1: Frequent Purchasers - Credit Limit Increase
- CL2: Low Engagement - Re-engagement Campaigns
- CL3: Cash Advance Dependent - Risk Mitigation
- CL4: High Value Spenders - High Tier Rewards

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LIMITATIONS

- Dataset and Scope
- Imputation Strategy
- PCA Variance Explained.
- Interpretability of Components